Handover Probability in Drone Cellular Networks

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Project Domain & Objectives

- Analyzes a cellular network where mobile Drone Base Stations (DBSs) serve a stationary ground user.
- **Objective:** Evaluate how drone mobility and parameter variations trigger network handovers over time.

The Role of Uncertainty

- Real-world drone networks are highly dynamic; parameters like drone density, operating speeds, and distances change with every deployment.
- Relying on a purely deterministic model only confirms mathematical theory at one specific set of values.
- Incorporating uncertainty ensures our system models realistic, diverse scenarios rather than a single idealized state.

Milestone 3 Recap

The M3 Deterministic Setup

- Modeled DBS initial positions using a Poisson Point Process (PPP).
- **Same Speed Model (SSM):** All drones moved in straight lines at a fixed, constant speed.
- **Fixed Parameters:** Density (1.0×10^{-6} DBS/m^2), Speed (45 km/h), Exclusion radius (500 m), and Time window (400 s).

Limitations of the Deterministic Approach

- A deterministic baseline only tests one specific, hand-picked setting.
- It fails to prove if the underlying stochastic geometry theory holds true across the natural variations of real-world drone deployments.

Simulation Architecture

End-to-End Pipeline

- **Initialization:** `randomised_config.py` dynamically draws physical parameters.
- **Simulation:** Monte Carlo realizations track drone movements and connections.
- **Theory:** Analytical scripts load identical parameters to compute exact and lower-bound probabilities.
- **Evaluation:** Outputs are plotted to compare SSM vs. DSM metrics.

The Decision Rule

Nearest-Neighbor Policy:
The user always connects to the physically closest drone at every moment in time.

$$x^*(t) = \arg\min_i ||x_i(t)||$$

A handover occurs when the index of this closest drone changes.

Why Determinism Falls Short

Key Sources of System Uncertainty

- **Spatial Distribution:** Drones land and start at varying distances from ground users.
- **Network Density:** The number of active drones deployed fluctuates per mission.
- **Drone Speeds:** Individual drones move at different speeds due to hardware differences, battery levels, or wind.

The Need for Randomization

By drawing parameters randomly, every simulation run becomes a genuinely different experiment. If the theoretical math matches the simulation across all these variations, we prove the results are robust and generalizable, not just lucky for one special case.

Distributional Assumptions

The `randomised_config.py` Module

Table: Sampled Distributions for Milestone 4

Random Variable	Distribution & Range
Drone Density (λ_0)	Uniform (5×10^{-7} to 2×10^{-6} UAVs/m^2)
Mean Speed (v_{Mean})	Uniform (30 to 60 km/h)
Speed Spread (v_{Diff})	Uniform (20 to 120 km/h)
Exclusion Radius (u_0)	Uniform (300 to 800 m)
Time Window (T)	Uniform (200 to 400 s - integer)

Distributional Modeling:

- Initial drone locations follow a Poisson Point Process (PPP).
- **Different Speed Model (DSM):** Individual speeds (v) are drawn from a Rayleigh or Uniform distribution.

Introducing Parameter Variability

Design and Decision Component

- Randomness is introduced via a central module called `randomised_config.py`.
- This module draws five key physical parameters from Uniform distributions for each run.
- The nearest-neighbor decision rule, handover detection logic, and straight-line drone motion remain exactly as they were in the baseline.

Table: Parameter Comparison: Deterministic vs. Randomized

Parameter	M3 Deterministic (Fixed)	M4 Randomized (Actual Draw)
Density (λ_0)	$1.00 \times 10^{-6}/\text{m}^2$	$1.44 \times 10^{-6}/\text{m}^2$
Mean Speed	45.0 km/h	53.1 km/h
Exclusion Radius (u_0)	500 m	335.8 m
Time Window (T)	400 s	272 s

The Anchor Drone Concept

Addressing Uncertainty

- Testing across many different randomly drawn settings provides a much more convincing and robust conclusion than testing a single hand-picked set of values.

Why Randomization Improves System Stability

- **Deterministic SSM:** All drones move at the exact same speed. The swarm shifts uniformly, forcing rapid, unavoidable handovers.
- **Randomized DSM:** Each drone gets a different speed drawn from a distribution.
- **The Result:** Drones that randomly draw slow speeds act as natural "anchors." They stay closest to the user for a much longer time before a faster drone can overtake them, naturally filtering for connection stability.

Simulation Pipeline Workflow

Integration Architecture

- A save-and-load system guarantees the simulation and theoretical scripts evaluate the exact same randomized configurations.

Execution Workflow

1. **Sample:** `handover_dbs_simulation.py` calls `sample_params()` to draw the variables.
2. **Save:** Parameters are saved immediately to a local JSON file.
3. **Simulate:** 10,000 Monte Carlo realizations run using the sampled density, speeds, and time vectors.
4. **Validate:** Theory scripts read the exact same JSON file to compute the analytical integrals.
5. **Evaluate:** `plot_handover.py` aggregates the outputs and generates comparative figures.

Domain-Specific Indicators

- **Handover Probability $P[H(t)]$:** The foundational metric. Measures the probability that a ground user has had to switch their drone connection at least once by time t.
- **Expected Communication Cost:** Measures resource utilization. A lower handover probability directly means less signaling overhead (authentication, resource allocation) and better bandwidth preservation.
- **Accumulated Latency:** Measures user experience via service interruption. Each handover causes an approximate 40 ms interruption; delaying handovers reduces this latency.
- **Connection Stability $CS(t)$:** Defined mathematically as $1 - P[H(t)]$. It measures the fraction of users who have *not* experienced any handover at time t.

Handover Probability: SSM vs DSM

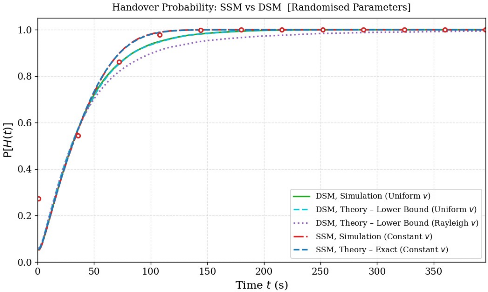

Figure: Handover Probability over Time

Performance Comparison

- DSM reaches 90% handover probability at 104 s, while SSM reaches it at only 60 s.
- This 44-second improvement means randomized DSM users stay on their original drone significantly longer.
- **Resource Savings:** The peak communication cost saving is 10.8% at $t \approx 88$ s for the DSM approach.

Conclusions and Remaining Challenges

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Main Observations

- **Theory Robustness:** The mathematical models are highly accurate even under randomized parameters (MSE of 6.2×10^{-6}).
- **Density Impact:** Higher drone densities speed up handovers due to increased competition for the nearest-neighbor position.
- **DSM Superiority:** Variable-speed drones clearly provide better connection stability for users needing longer sessions.

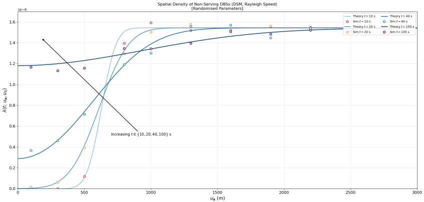

Figure: Spatial Density over Time

Remaining Challenges

- Drones drifting outside the finite 30 km simulation boundary slightly reduce effective density at late times, causing minor theory/simulation mismatches.